Customer Attrition Classification

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Introduction

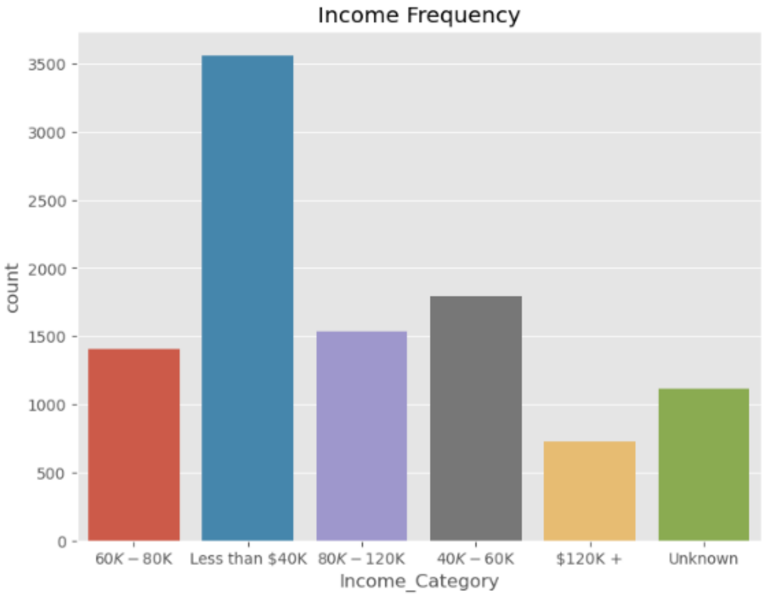
- Customer attrition is important for all types of businesses
- Banks and credit card companies specifically have an increased desire to keep their customers loyal
- The goal of this project is to build a model that can correctly identify and flag customers about to churn
- Using the results would allow the company can make efforts to retain their loyalty

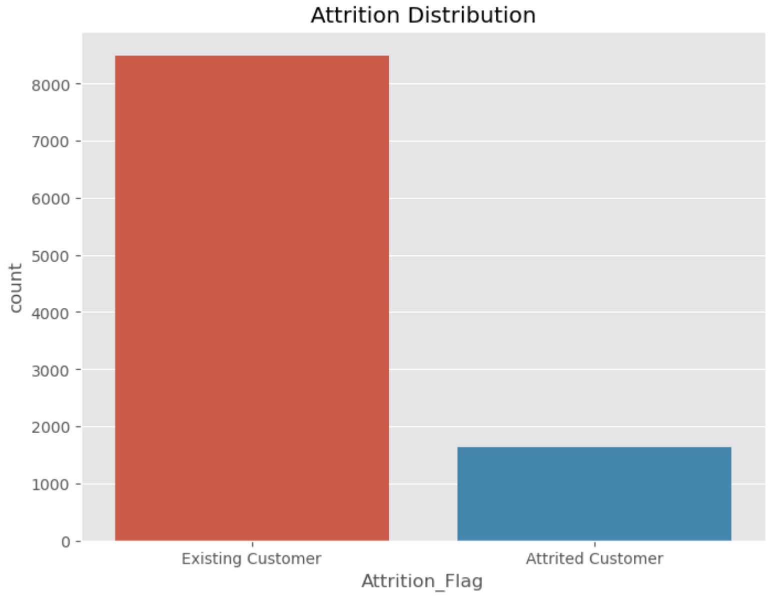
Data Description

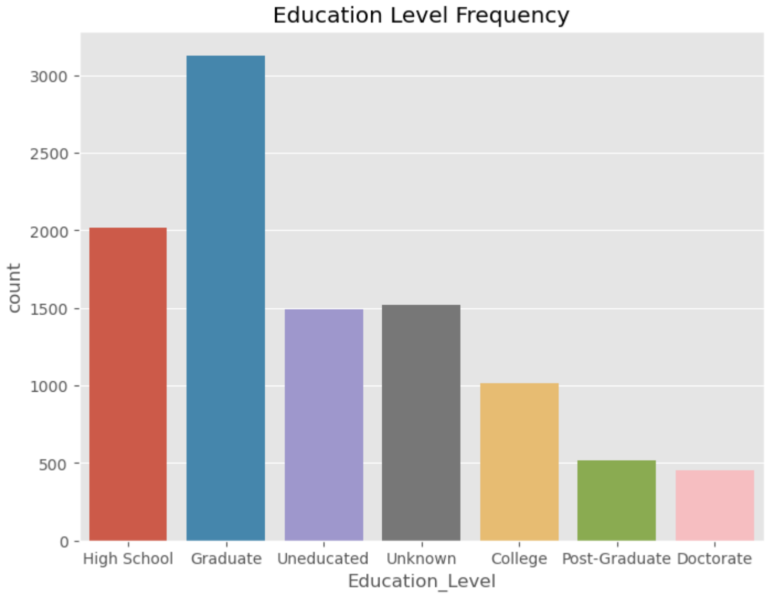
- This Dataset was found on Kaggle
- It contains 10,126 rows and 23 columns with no missing values

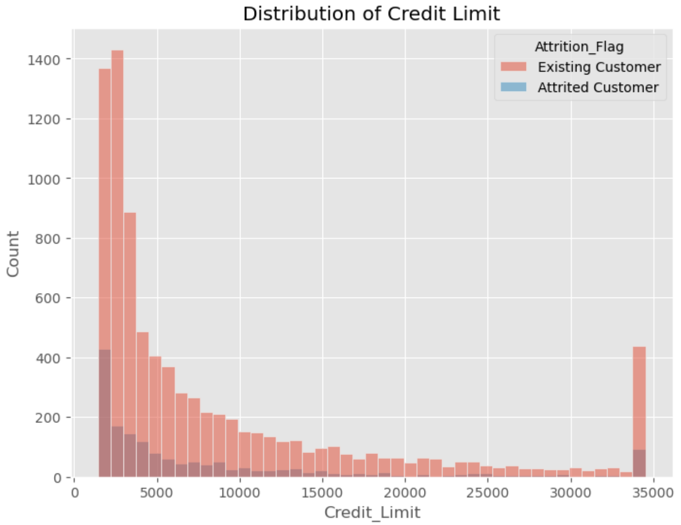
Column name	Type	Description
CLIENTNUM	Numerical	Unique identifier for each customer.
Attrition_Flag	Categorical	Flag indicating whether or not the customer has churned out.
Customer_Age	Numerical	Age of customer.
Gender	Categorical	Gender of customer.
Dependent_count	Numerical	Number of dependents that customer has.
Education_Level	Categorical	Education level of customer.
Marital_Status	Categorical	Marital status of customer.
Income_Category	Categorical	Income category of customer.
Card_Category	Categorical	Type of card held by customer.
Months_on_book	Numerical	How long customer has been on the books.
Total_Relationship_Count	Numerical	Total number of relationships customer has with the credit card provider.
Months_Inactive_12_mon	Numerical	Number of months customer has been inactive in the last twelve months.
Contacts_Count_12_mon	Numerical	Number of contacts customer has had in the last twelve months.
Credit_Limit	Numerical	Credit limit of customer.
Total_Revolving_Bal	Numerical	Total revolving balance of customer.
Avg_Open_To_Buy	Numerical	Average open to buy ratio of customer.
Total_Amt_Chng_Q4_Q1	Numerical	Total amount changed from quarter 4 to quarter 1.
Total_Trans_Amt	Numerical	Total transaction amount.
Total_Trans_Ct	Numerical	Total transaction count.
Total_Ct_Chng_Q4_Q1	Numerical	Total count changed from quarter 4 to quarter 1.
Avg_Utilization_Ratio	Numerical	Average utilization ratio of customer.

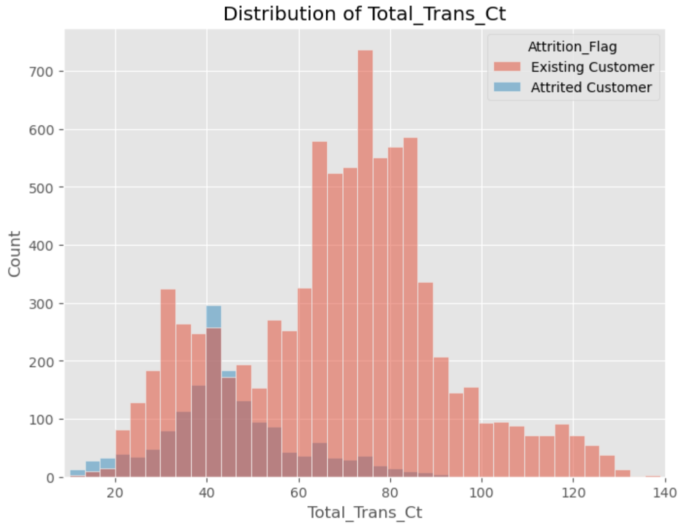
Exploratory Data Analysis

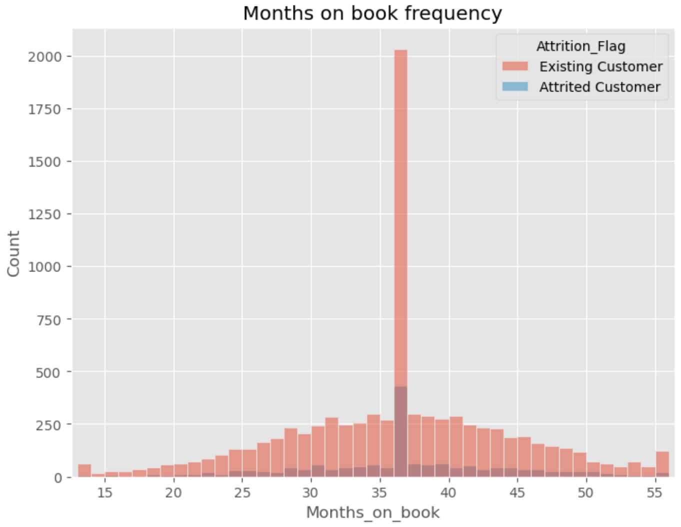


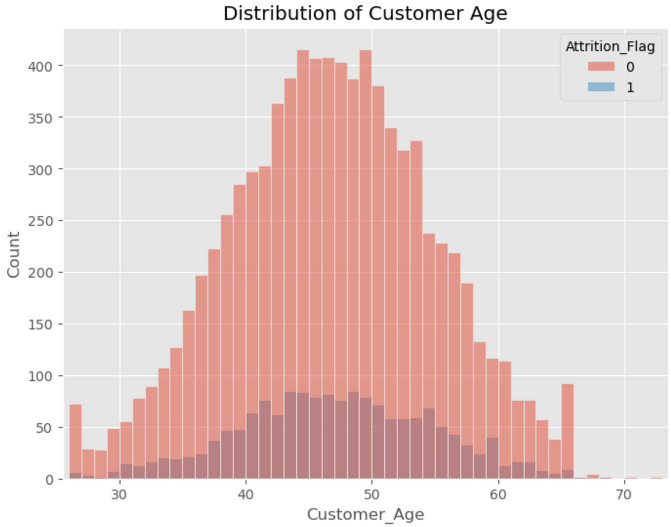


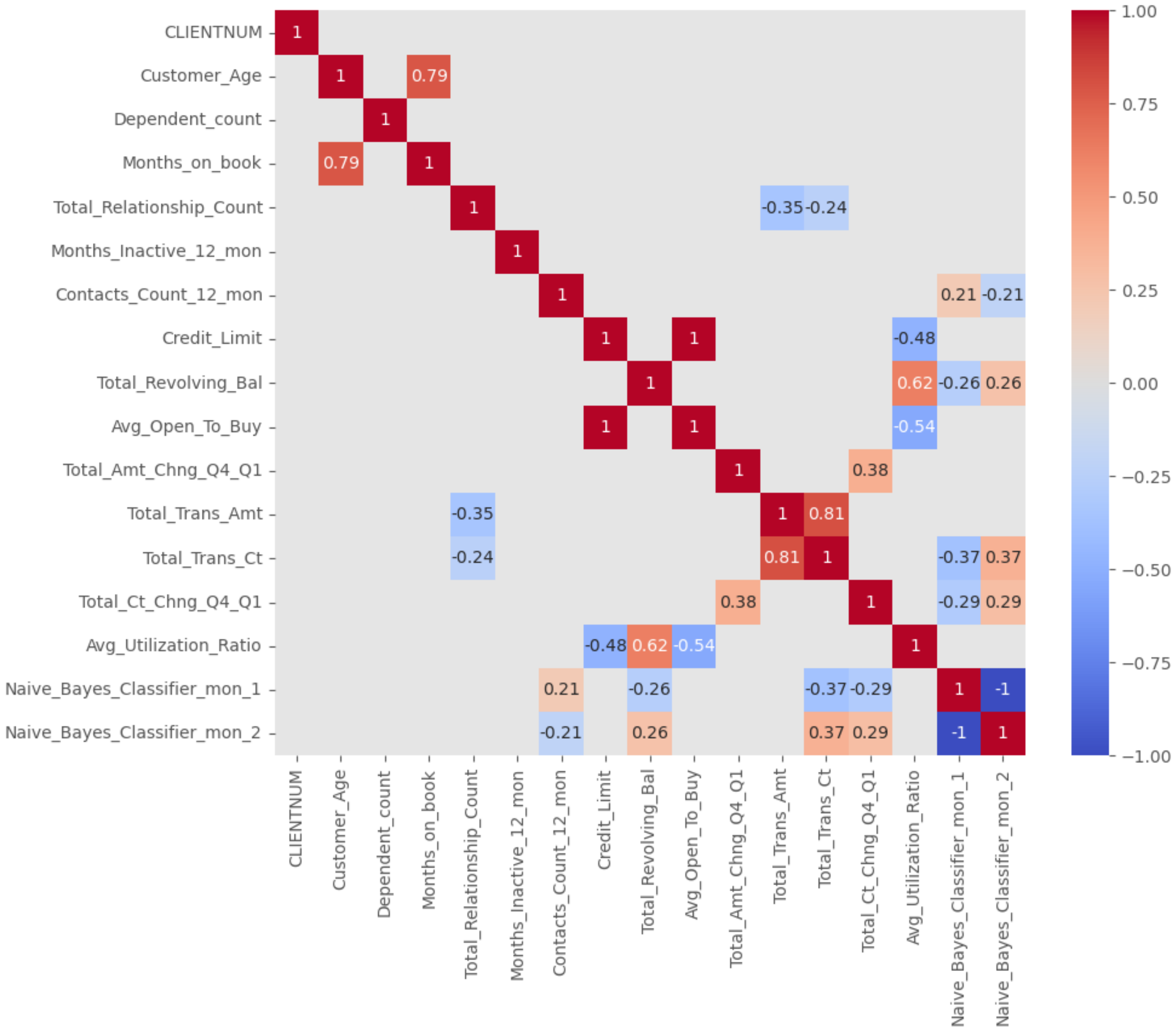



Distribution of Credit Limit
Attrition_Flag
Existing Customer
Attrited Customer
Count
Credit_Limit
0
200
400
600
800
1000
1200
1400
5000
10000
15000
20000
25000
30000
35000


Distribution of Total_Trans_Ct
Attrition_Flag
Existing Customer
Attrited Customer
Count
Total_Trans_Ct
0
100
200
300
400
500
600
700
20
40
60
80
100
120
140


Months on book frequency
Attrition_Flag
Existing Customer
Attrited Customer
Count
Months_on_book
0
250
500
750
1000
1250
1500
1750
2000
15
20
25
30
35
40
45
50
55


Distribution of Customer Age
Attrition_Flag
0
1
Count
Customer_Age
0
50
100
150
200
250
300
350
400
30
40
50
60
70


CLIENTNUM
Customer_Age
Dependent_count
Months_on_book
Total_Relationship_Count
Months_Inactive_12_mon
Contacts_Count_12_mon
Credit_Limit
Total_Revolving_Bal
Avg_Open_To_Buy
Total_Amt_Chng_Q4_Q1
Total_Trans_Amt
Total_Trans_Ct
Total_Ct_Chng_Q4_Q1
Avg_Utilization_Ratio
Naive_Bayes_Classifier_mon_1
Naive_Bayes_Classifier_mon_2
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Modeling

Feature Engineering

- We made interaction terms based on variables which had high correlations
- Along with flags based off our visualizations

Column name	Type	Description
Interaction_1	Numerical	Customer_Age * Months_on_book
Interaction_2	Numerical	Total_Revolving_Bal * Avg_Utilization_Ratio
Interaction_3	Numerical	Total_Trans_Amt * Total_Trans_Ct
Interaction_4	Numerical	Avg_Utilization_Ratio * Avg_Open_To_Buy
Months_on_book_36_flag	Categorical	Flag for Months_on_book = 36
Credit_Limit_max_flag	Categorical	Flag for Credit_Limit = 34516
Total_Revolving_Bal_0_flag	Categorical	Flag for Total_Revolving_Bal = 0
Total_Revolving_Bal_max_flag	Categorical	Flag for Total_Revolving_Bal = 2517
Total_Trans_Ct_under_60	Categorical	Flag for Total_Trans_Ct >= 60

Feature Selection

- Used RFECV with 5 folds and f1 score as the evaluation metric
- We ran through this sequence 5 times and kept track of the percentage of time each feature was selected in each model

Model	Number of features
Random Forest	12
Ada Boost	20
Gradient Boosting	15
Hist Gradient Boost	14
XG Boost	13
Light GBM	22
Cat Boost	25

Hyper-Parameter Tuning

- Used the Optuna framework to maximize the f1 score over 50 trials

Model	Hyper-Parameters
Random Forest	n_estimators = 974 max_depth = 10 min_samples_split = 5 min_samples_leaf = 2
Ada Boost	learning_rate = 0.08 n_estimators = 438 max_depth = 3
Gradient Boosting	n_estimators = 958 max_depth = 3 min_samples_split = 8 min_samples_leaf = 6 learning_rate = 0.0547
Hist Gradient Boost	max_iter = 520 learning_rate = 0.1 l2_regularization = 0.1 min_samples_leaf = 7 max_leaf_nodes = 103 max_depth = 3 max_bins = 205
XG Boost	learning_rate = 0.02 n_estimators = 622 max_depth = 6 min_child_weight = 1 gamma = 0.003011 alpha = 3.305498 colsample_bytree = 0.51 subsample = 0.433741
Light GBM	n_estimators = 966 learning_rate = 0.02 num_leaves = 313 min_data_in_leaf = 37 min_child_weight = 0.07766 max_depth = 9 bagging_fraction = 0.65689 feature_fraction = 0.52445 lambda_l1 = 0.9316 lambda_l2 = 1.0916
Cat Boost	iterations = 699 learning_rate = 0.1 min_data_in_leaf = 154 depth = 7 random_strength = 0.76723 bagging_temperature = 0.34078 border_count = 79 l2_leaf_reg = 64

Model Evaluation

Model	Accuracy score
Random Forest	.9637
Ada Boost	.9744
Gradient Boosting	.9736
Hist Gradient Boosting	.9734
XG Boost	.9716
Light GBM	.9749
Cat Boost	.9763

Ensemble Model

- After identifying the optimal features and hyper-parameters for our dataset we picked out our top 5 performing models by accuracy score to ensemble
- Final model scored an accuracy of .98 and a recall score of .95

Conclusion and Further Study

- With the high level of accuracy our model was able to produce, we believe this model would be very beneficial to credit card companies or companies where customer attrition is a main priority
- An area of further study we would be interested in looking at is specific causes customers to churn from the company if the data is available
- Our dataset focuses on customers’ personal data but does not have any information on what customers could be dissatisfied with